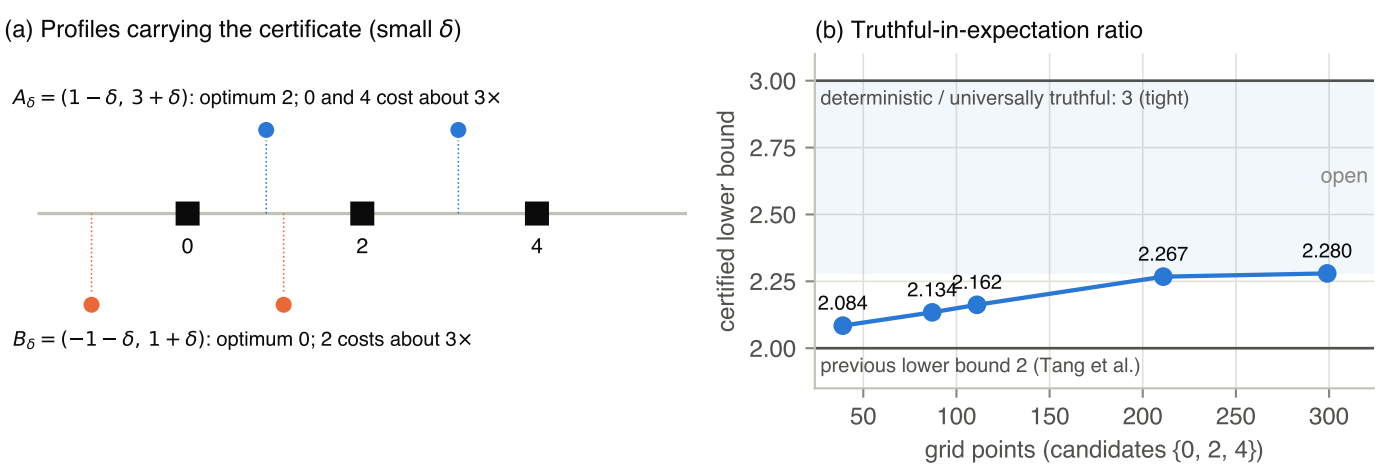
(a) Profiles carrying the certificate (small δ)
$A_\delta = (1-\delta,\ 3+\delta)$: optimum 2; 0 and 4 cost about 3×
0
2
4
$B_\delta = (-1-\delta,\ 1+\delta)$: optimum 0; 2 costs about 3×
(b) Truthful-in-expectation ratio
3.00
2.75
2.50
2.25
2.00
certified lower bound
deterministic / universally truthful: 3 (tight)
open
previous lower bound 2 (Tang et al.)
2.084
2.134
2.162
2.267
2.280
50
100
150
200
250
300
grid points (candidates {0, 2, 4})